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Author(s): Roy Penchansky and J. William Thomas

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The Concept of Access

Definition and Relationship to Consumer Satisfaction

ROY PENCHANSKY, D.B.A.,* AND J. WILLIAM THOMAS, PH.D.†

Access is an important concept in health policy and health services research, yet it is one which has not been defined or employed precisely. To some authors "access" refers to entry into or use of the health care system, while to others it characterizes factors influencing entry or use. The purpose of this article is to propose a taxonomic definition of "access." Access is presented here as a general concept that summarizes a set of more specific dimensions describing the fit between the patient and the health care system. The specific dimensions are availability, accessibility, accommodation, affordability and acceptability. Using interview data on patient satisfaction, the discriminant validity of these dimensions is investigated. Results provide strong support for the view that differentiation does exist among the five areas and that the measures do relate to the phenomena with which they are identified.

WHILE "access" is a major concern in health care policy and is one of the most frequently used words in discussions of the health care system, most authorities agree that it is not a well-defined term.¹⁻³ For example, Aday and Anderson state, "Just what the concept of access means . . . [is] ill-defined at present. Thus far, access has been more of a political than an operational idea . . . few attempts have been made to provide formalized conceptual or empirical definitions of access."¹ The problem is

not limited to the lack of a precise definition for access, or the multiple meanings given to the term; access also is used synonymously with such terms as accessible and available, which are themselves ill-defined. *The Discursive Dictionary of Health Care*, published by the U.S. House of Representatives, should be a source of precise definitions for terms employed in federal health care legislation. However, the definition for access states that the term ". . . is thus very difficult to define and measure operationally . . ." and that ". . . access, availability and acceptability . . . are hard to differentiate."⁴

A few authors equate access with entry into or use of the system; examples are ". . . the first barrier to access . . ."⁵ or ". . . access refers to entry into."⁶ While access is more often employed to characterize factors which influence entry or use, opinions differ concerning the range of factors included within access and whether access is seen as characterizing the resources or the clients. These variations can be seen in the

* Professor, School of Public Health, University of Michigan.

† Assistant Professor, School of Public Health, University of Michigan.

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From the Department of Medical Care Organization, School of Public Health, University of Michigan.

Address for reprints: Roy Penchansky, Department of Medical Care Organization, School of Public Health, University of Michigan, 109 Observatory Street, Ann Arbor, MI 48109.

different interpretations of the public policy goal of "equal access."⁷ Some assume that this means the guarantee of availability, supply and resources⁸; while to others it means insuring equal use for equal need.^{2,9} The first view focuses on the system having attributes that allow entry or use if desired and suggests that access deals with only the limited set of such attributes. The second interpretation suggests that access encompasses all factors that influence the level of use, given a health care need. The use of access as a construct measured by the discrepancy between entry or use and need has contributed further to confusion about the dimensions included in the term.¹⁻³

It is clear that access is most frequently viewed as a concept that somehow relates to consumers' ability or willingness to enter into the health care system. The need for such a concept derives from the repeated observation that entry into (or use of) the health care system cannot be fully explained by analyzing the health state of clients or even their general concerns with health care. If there are phenomena beyond these which significantly influence the use of health care services, then these phenomena should be defined and measured. This information could then be used to influence the system in a manner to obtain desired intermediate or final outcomes.

The purpose of this article is to propose and test the validity of a taxonomic definition of access, one that disaggregates the broad and ambiguous concept into a set of dimensions that can be given specific definitions and for which operational measures might be developed. In the following section, these dimensions of access are defined and related to previous references to access in literature dealing with health services utilization. The proposed dimensions are then related to research findings on patient satisfaction. Next, using interview data from Rochester, New York, the discriminant validity of the dimensions is

tested through a factor analysis of responses to questions concerning satisfaction with various characteristics of health services and providers. Finally, regression analyses are performed on the data to investigate construct validity of measures of the dimensions, with the measures serving as dependent variables in the regression equations.

Access Defined

"Access" is defined here as a concept representing the degree of "fit" between the clients and the system. It is related to—but not identical with—the enabling variables in the Anderson¹⁰ model of the determinants of use, a model which includes variables describing need, predisposing factors and enabling factors. Access is viewed as the general concept which summarizes a set of more specific areas of fit between the patient and the health care system. The specific areas, the dimensions of access, are as follows:

Availability, the relationship of the volume and type of existing services (and resources) to the clients' volume and types of needs. It refers to the adequacy of the supply of physicians, dentists and other providers; of facilities such as clinics and hospitals; and of specialized programs and services such as mental health and emergency care.

Accessibility, the relationship between the location of supply and the location of clients, taking account of client transportation resources and travel time, distance and cost.

Accommodation, the relationship between the manner in which the supply resources are organized to accept clients (including appointment systems, hours of operation, walk-in facilities, telephone services) and the clients' ability to accommodate to these factors and the clients' perception of their appropriateness.

Affordability, the relationship of prices of services and providers' insurance or deposit requirements to the clients' income, ability to pay, and existing health insurance. Client perception of worth relative to total cost is a concern here, as is clients'

knowledge of prices, total cost and possible credit arrangements.

Acceptability, the relationship of clients' attitudes about personal and practice characteristics of providers to the actual characteristics of existing providers, as well as to provider attitudes about acceptable personal characteristics of clients. In the literature, the term appears to be used most often to refer to specific consumer reaction to such provider attributes as age, sex, ethnicity, type of facility, neighborhood of facility, or religious affiliation of facility or provider. In turn, providers have attitudes about the preferred attributes of clients or their financing mechanisms. Providers either may be unwilling to serve certain types of clients (e.g., welfare patients) or, through accommodation, make themselves more or less available.

Concepts embodied in these dimensions have been identified previously in the literature.† Access is clearly identified with *affordability* by Bice et al.,¹² when they say "... Medicare and Medicaid have probably played a major role in increasing access," and in their references to prices of services and income. **Although they define access in terms of client socioeconomic factors,** Bice et al. also mention distance traveled (*accessibility*), and "relative lack of supply or *availability*," Donabedian uses the phrase "socio-organizational accessibility" and gives examples "... the reluctance of some men to see a woman physician and the refusal of some white dentists to treat black patients."¹³ This we call *acceptability*. He also presents the concept of geographical *accessibility* which, he indicates, deals with the location of service and the impact of consumer travel time, distance, cost and effort on use. *Availability* is used by Donabedian to refer to the service-producing capacity of resources, which is the supply side of the supply/demand relationship in our definition of the term.

Fein discusses access in terms of the determinants of the allocation of services, with the outcomes being the supply of ser-

vices, by type and geographic area, for a specific clientele.¹⁴ This is *availability*. He also emphasized personal income—a key to *affordability*—as a major determinant of access.

In providing criteria for accessibility Freeborn and Greenlick appear to be referring to a number of dimensions—*accessibility*, *accommodation*, and *availability*—when they say that "... individuals should have access to the system at the time and place needed, through a well-defined and known point of entry. A comprehensive range of personnel, facilities and services that are known and convenient should be available."¹⁵ Simon et al. describe an "index of accessibility" for measuring the timeliness of response to patients' requests to enter the system, the appropriateness of the response to patients' requests to enter the system, and the effort (in terms of time spent) that the patient must expend to be served.¹⁶ Two access dimensions seem to be addressed: *availability*, relating to timeliness of the response, and *accommodation*, relating to patient time spent in being served.

Clearly, the dimensions of access are not easily separated. In some settings accessibility may be closely tied to availability. Yet, various service areas having equivalent availability may have different accessibility. **In explaining where persons actually go for care, the more important dimension (within some parameter of accessibility) is often acceptability and not accessibility.**¹⁷⁻¹⁹ **Availability undoubtedly affects accommodation and acceptability.** When the level of demand is high relative to supply, physicians practice in different ways and have differing ability to select the clients they desire to serve. The five dimensions surely represent closely related phenomena, which explains why they have been seen as part of a single concept: access. At issue is whether they are sufficiently distinct to be measured and studied separately.

† For a more thorough review of literature related to the concept of access, see Penchansky.¹¹

Satisfaction With Access

Problems with access, or more specifically with any of the component dimensions of access, are presumed to influence clients and the system in three measurable ways: 1) **utilization of services**, particularly entry use, will be lower, other things being equal; 2) **clients will be less satisfied** with the system and/or the services they receive; and 3) **provider practice patterns may be affected** (such as when inadequate supply resources cause physicians to curtail preventive services, devote less than appropriate amounts of time to each of their patients or use the hospital as a substitute for their short supply). While it is necessary to examine the concept of access in terms of all of these effects, we shall focus here only on the second: patient satisfaction. A subsequent paper will explore the relationship of the definitions presented to utilization of ambulatory services.

In some satisfaction studies, researchers have employed general measures of patient satisfaction,²⁰⁻²⁴ but in other cases measures focusing on specific aspects of patient/system relationships have been used. For example, Hulka and her colleagues have in several studies investigated factors associated with patient attitudes toward providers' technical competence, providers' personal qualities, and the costs/convenience of getting care.²⁵⁻²⁸ In an excellent review of patient satisfaction literature, Ware et al.²⁹ defined eight dimensions of patient satisfaction that have been addressed in published studies: art of care (encompassing, for example, personal qualities), technical quality of care (relating to provider professional competence), accessibility/convenience, finances, physical environment, availability, continuity and efficacy/outcomes of care.

Appropriately, several of these dimensions of patient satisfaction are identical or closely related to the access dimensions defined above. "Availability" refers to the same concept in our access taxonomy and

in Ware et al.'s satisfaction taxonomy, and "finances," as defined by Ware et al., is essentially the same as *affordability*. The accessibility/convenience dimension described by Ware et al. is a composite of two access dimensions: *accessibility* and *accommodation*. While *acceptability* in the access taxonomy encompasses factors that Ware et al. group under "physical environment," acceptability is a broader concept that also includes patient attitudes toward provider personal characteristics as well as toward other characteristics of the provider's practice.

Ware et al. note that although researchers have constructed measurement scales focusing on various dimensions of patient satisfaction, the ability of these scales to distinguish among different aspects of satisfaction has not been shown:

For example, can measures distinguish between satisfaction with financial aspects of care . . . and with art of care? . . . the discriminant validity of satisfaction scores must be demonstrated and well understood before they are used to make judgments about specific characteristics of providers and services. Findings published to date do not justify the use of patient satisfaction ratings for this purpose.²⁹

In the next section we address this issue and investigate discriminant and construct validity of the proposed access dimensions and their related measures.

Methods and Results

Source of Data

Data used for this study were obtained from a survey conducted in Rochester, New York in 1974. The principal purpose of the survey was to investigate factors, including satisfaction with existing sources of care, that influence respondents' choice of health care plan. The survey population consisted of hourly employees of a General Motors Corporation electrical parts assembly plant and their spouses. Two questionnaires were used: one for employees, which included questions concerning family financial status, health care expendi-

TABLE 1. Patient Satisfaction Questions

Access Dimension	Questions
AVAILABILITY	1. All things considered, how much confidence do you have in being able to get good medical care for you and your family when you need it? 2. How satisfied are you with your ability to find one good doctor to treat the whole family? 3. How satisfied are you with your knowledge of where to get health care? 4. How satisfied are you with your ability to get medical care in an emergency?
ACCESSIBILITY	5. How satisfied are you with how convenient your physician's offices are to your home? 6. How difficult is it for you to get to your physician's office?
ACCOMMODATION	7. How satisfied are you with how long you have to wait to get an appointment? 8. How satisfied are you with how convenient physicians' office hours are? 9. How satisfied are you with how long you have to wait in the waiting room? 10. How satisfied are you with how easy it is to get in touch with your physician(s)?
AFFORDABILITY	11. How satisfied are you with your health insurance? 12. How satisfied are you with the doctors' prices? 13. How satisfied are you with how soon you need to pay the bill?
ACCEPTABILITY	14. How satisfied are you with the appearance of the doctor's offices? 15. How satisfied are you with the neighborhoods their offices are in? 16. How satisfied are you with the other patients you usually see at the doctors' offices?

tures, hospital experience, health insurance coverage, personal health problems and use of medical care services; and one for spouses, which asked about sources of care, personal health problems, use of medical care services and satisfaction with various characteristics of providers and the medical care system. A response rate of 83 per cent was achieved, yielding 626 completed or partially completed employee questionnaires. However, since satisfaction questions were addressed only to spouses, the sample for the current study included 287 people who completed all satisfaction questions in the survey.

Construction of Satisfaction Scales

Responses to questions concerning satisfaction with various characteristics of the medical care system and the patient's usual

provider were scored on a five-point Likert scale, ranging from "very satisfied" to "very dissatisfied." As shown in Table 1, 16 of the satisfaction items were hypothesized to relate to specific dimensions of access: four to availability, two to accessibility, four to accommodation and three each to affordability and acceptability. Respondent satisfaction with each of the access dimensions was determined using the method of summated ratings,³⁰ and ranges of the summated ratings were standardized to zero (very satisfied) to one (very dissatisfied). Distributions for these summated ratings of satisfaction are presented in Figure 1. Consistent with findings in other studies, respondents appear to be generally satisfied with all dimensions of access. Proportions of respondents who are relatively dissatisfied, scoring 0.75 or higher, total only 5.0 per cent for availability, 7.3

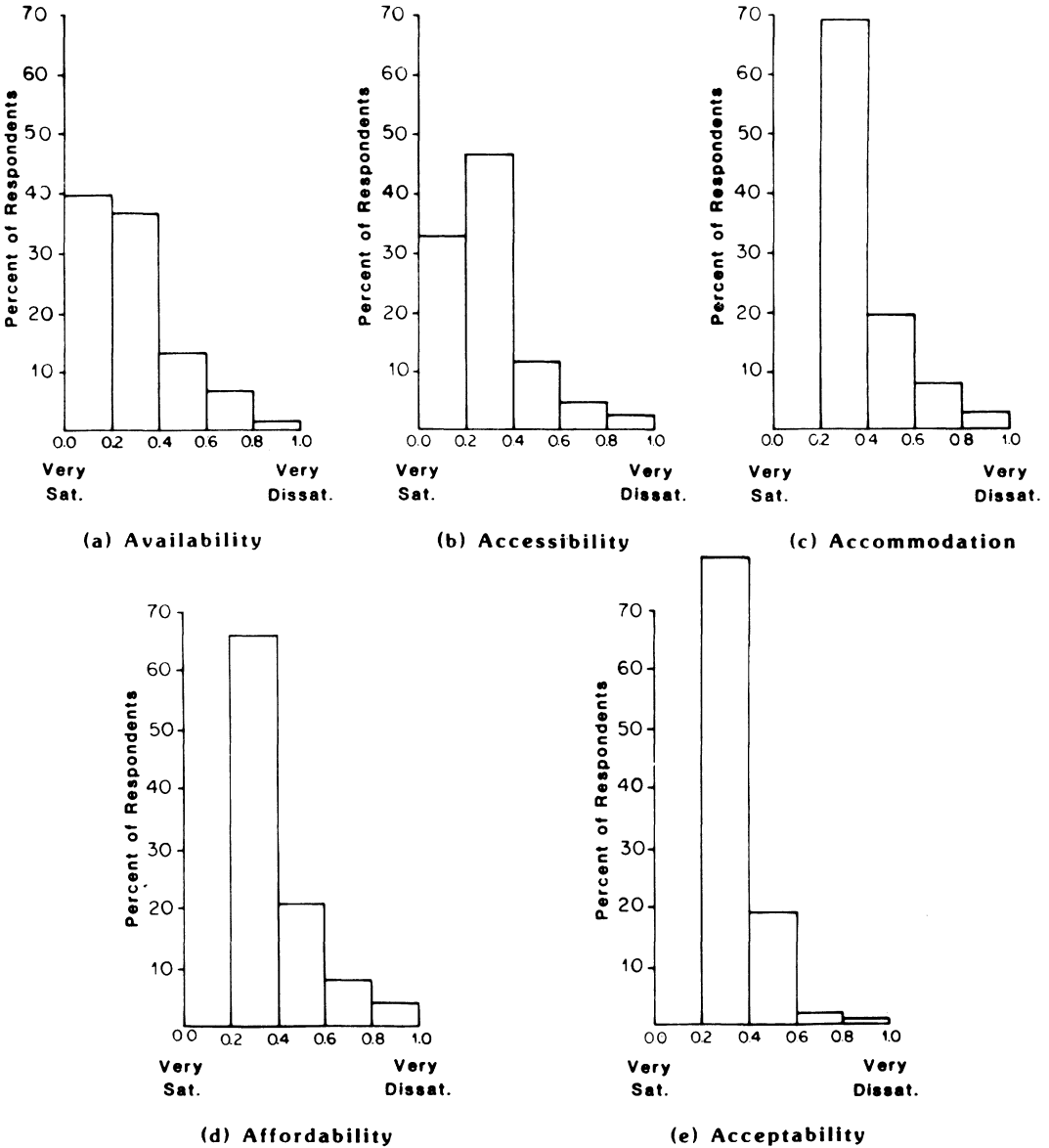


FIG. 1. Distributions of Summated Measures of Access Dimensions.

per cent for accessibility, 6.2 per cent for accommodation, 7.7 per cent to affordability and 1.4 per cent for acceptability. While none of the respondents was highly satisfied (scoring 0.2 or lower) with accommodation, affordability or acceptability, a substantial majority indicated general satisfaction (scoring 0.2 to 0.4) with these dimensions.

Discriminant Validity

To establish discriminant validity it is necessary to show that respondent's perceptions of the proposed dimensions are independent and that relationships between specific satisfaction items and the dimensions of access are as hypothesized. The degree to which phenomena as-

TABLE 2. Correlations (Goodman-Kruskal Gamma) Among Summated Ratings

	Availability	Accessibility	Accommodation	Affordability	Acceptability
Availability	1.0	.227	.379	.370	.359
Accessibility	—	1.0	.349	.330	.274
Accommodation	—	—	1.0	.469	.415
Affordability	—	—	—	1.0	.436
Acceptability	—	—	—	—	1.0

sociated with the five dimensions are perceived independently by respondents is indicated in Table 2 by correlations among the summated ratings. Although respondents expressed high levels of satisfaction with all dimensions, their perceptions of the dimensions appear to be generally independent.

To provide a more rigorous test of discriminant validity a factor analysis was performed on the sixteen access-related satisfaction items listed in Table 1. Using the principal axis method³¹ with 0.25 specified as the minimum eigenvalue, the analysis yielded five factors, which then were rotated simultaneously using the varimax algorithm.³² Table 3 shows loadings of individual items on the rotated factors. Each of the first three factors explained almost 12 per cent of the item variance while factors four and five explained only 7 per cent and 5 per cent of the variance, respectively.

As seen in Table 3, the four “accommodation” items have the largest positive loadings on the first factor, and the four “availability” items have the largest positive loadings on the second factor. The three “acceptability” items load highest on the third factor and the two “accessibility” items highest on factor 4. Two of the three “affordability” items, doctor’s prices and how soon you need to pay the bill, load highest on factor 5. The other “affordability” item, satisfaction with your health insurance, does not load highly on any factor. There was little variability in responses to this question (fewer than three per cent of respondents indicated any dissatisfaction

with their coverage), perhaps because all members of the study population share similar health insurance benefits.

Each of the five factors was labeled according to the related access dimension. Table 4, which presents Goodman-Kruskal gammas³³ for all pairs of factors and summated measures, indicates strong association between each related factor and summated measure and a low degree of association between unrelated pairs. Thus it appears that, for the population included in this study, differentiation does exist among the five proposed access dimensions and that both factor scores and summated ratings are satisfactory measures for these dimensions.

Construct Validity

As noted by Ware et al.,²⁹ one test of validity “. . . is whether measures of specific satisfaction dimensions differentiate between specific characteristics of providers and medical care services,”; that is, are these dimensions valid in terms of the phenomena to which they are supposed to relate? Thus one would expect travel time to correlate more highly to satisfaction with accessibility than to acceptability, and that waiting time for an appointment would be a more important correlate of satisfaction with accommodation than affordability. In order to investigate this aspect of validity, five least squares regressions were performed to relate the set of independent variables shown in Table 5 to each of the factor measures de-

TABLE 3. Factor Loadings for Satisfaction Items

Dimension	Items (satisfaction with:)	Factors				
		(1)	(2)	(3)	(4)	(5)
Availability	1. Get Medical Care When You Need It	.097	.566	.061	-.170	.131
	2. Find One Good Doctor	.255	.647	.126	.168	.028
	3. Knowledge Of Where To Get Care	.106	.805	.132	.067	.119
	4. Get Emergency Care	.277	.444	.187	-.009	-.008
Accessibility	5. Convenient Location Of Offices	.245	.103	.186	.645	.222
	6. Difficulty In Getting To Office	.098	.018	-.028	.648	.050
Accommodation	7. Wait For Appointment	.765	.091	.066	.069	.196
	8. Convenience Of Office Hours	.576	.118	.134	.349	.140
	9. Wait In Waiting Room	.575	.217	.130	.151	.108
	10. Getting In Touch With Physician	.495	.289	.164	.116	.098
Affordability	11. Health Insurance	.214	.146	.058	.027	.103
	12. Doctor's Prices	.309	.142	.084	.168	.631
	13. How Soon To Pay Bill	.197	.127	.298	.147	.523
Acceptability	14. Appearance Of Offices	.184	.254	.626	.070	-.045
	15. Neighborhoods Offices Are In	.083	.080	.658	.020	.211
	16. Patients You Usually See There	.108	.100	.739	.049	.126
	% VARIANCE	12.2	24.0	34.3	41.1	46.8

TABLE 4. Correlations (Goodman-Kruskal Gamma) Between Factors and Summated Measures

Summated Ratings	Factors				
	Availability	Accessibility	Accommodation	Affordability	Acceptability
Availability	.826	.073	.219	.048	.018
Accessibility	.071	.931	.149	.137	.055
Accommodation	.123	.144	.848	.144	.066
Affordability	.155	.116	.313	.810	.105
Acceptability	.087	.063	.191	.212	.968

fined in the factor analysis above.‡ For comparative purposes, the same set of independent variables was used in each equation. Presumably, the subsets of independent variables having significant regression coefficients should differ among the five dimensions, and those variables shown to relate to each dimension should be reasonable in terms of the definition of the dimension.

Among the variables in Table 5 are various patient sociodemographic characteristics which previous studies have shown to relate to satisfaction.^{25, 34, 35} Also included is utilization of services (number of visits), which studies indicate is positively correlated with patient satisfaction,²⁹ although direction of causality in this relationship remains open to question.^{23, 29} Dissatisfaction with waiting times in physicians offices/clinics has been noted by Deisher et al.³⁶ and Alpert et al.,³⁷ while Hulka et al.^{26, 27} show that having a regular physician and having a longer relationship with the physician are associated with higher levels of satisfaction.

‡ Regressions were also performed using the summated ratings as independent variables. As will be described, results of the two sets of analyses, one using factor scores and one using summated ratings, were generally consistent. Because factor score distributions were approximately normal and distributions of the summated ratings were not (see Figure 1), the discussion focuses on results of regressions in which factor scores served as dependent variables.

Also included among the independent variables are attitudinal measures describing perceived health status, health concerns and income adequacy, all of which were constructed from multiple items using the method of summated ratings. Health status is a composite score of nine Likert-scaled items, such as "satisfaction with the way you usually feel," "satisfaction with your resistance to illness" and "compared to other persons your age, how much health care do you need?" Health concern is a composite of two items which address how much the respondent thinks about his or her health; income adequacy is composed of two items, one asking about the adequacy of the respondent's income for meeting basic needs, and another asking whether he or she spends more or less than is earned. It was felt that perceived health status and health concerns might influence satisfaction with all of the dimensions of access, while income adequacy would relate only to affordability. Other independent variables such as "time to get an appointment" and "travel time to source of care" are also included because of hypothesized relationships with one or more dimensions of access.

The range of each independent variable was standardized between zero and one to facilitate interpretation of beta coefficients. Correlation coefficients calculated between pairs of independent variables were

TABLE 5. Independent Variables for Regression Equations

Variable		Distribution in Study Population	
1.	Race* (1 = white; 0 = black & other)	91%: White	9%: Black & other minorities
2.	Family Income	6%: ≤ \$10,000/yr. 37%: \$10,000–\$15,000/yr. 84%: ≤ 55 yrs.	25%: \$15,000–\$20,000/yr. 32%: > \$20,000/yr. 16%: > 55 yrs.
3.	Age* (1 = age ≤ 55 yrs; 0 = age > 55 yrs.)		
4.	Sex* (1 = male; 0 = female)	26%: Male	74%: Female
5.	Education* (1 = 12 yrs. or more; 0 = less than 12 yrs.)	77%: 12 yrs. or more	23%: Less than 12 yrs.
6.	Employment* (1 = housewife; 0 = other employed)	28%: Housewife	72%: Other employed
7.	Perceived Health Status (1 = poor)	30%: 0.0–0.2 46%: 0.2–0.4	17%: 0.4–0.6 5%: 0.6–0.8 2%: 0.8–1.0
8.	Health Concerns (1 = less concerned)	21%: 0.0–0.2 37%: 0.2–0.4	16%: 0.4–0.6 17%: 0.6–0.8 9%: 0.8–1.0
9.	Income Adequacy (1 = less adequate)	6%: 0.0–0.2 23%: 0.2–0.4	7%: 0.8–1.0
10.	Usual Source of Care* (1 = private physician; 0 = other)	87%: Private Physician	13%: Other
11.	Years with Usual Source* (1 = 2 yrs. or less; 0 = otherwise)	15%: 2 yrs. or less	85%: More than 2 yrs.
12.	Number of Sites Used By Family	2%: None 37%: One	5%: Four or more
13.	Number Family Ambulatory Visits During Last 6 Months	30%: 3 or less 33%: 4–7	8%: 18 or more
14.	Family Ambulatory Care Expenditures Last 6 Months	32%: \$50 or less 28%: \$50–\$100	16%: More than \$200
15.	Method of Travel to Care* (1 = own car; 0 = other)	93%: Own car	7%: Other
16.	Travel Time to Usual Source	50%: ≤ 15 min. 42%: 15–30 min.	7%: 30–45 min. 1%: More than 45 min.
17.	Time to Get Appointment	24%: Right away 33%: Couple of days	6%: A Month
18.	Wait Time in Physician's Office	38%: ≤ 15 min. 38%: 15–30 min.	13%: Couple of weeks 15%: 30–60 min. 8%: 60–90 min. 1%: > 90 min.

* Denotes binary variables.

TABLE 6. Beta Coefficients and R² Values for Regression Equations

Independent Variables	Regression (higher values indicate greater dissatisfaction)				
	Availability	Accessibility	Accommodation	Affordability	Acceptability
Constant	-.934	-1.460***	-.195	-1.152**	.485
Race (white = 1)	-.053	-.045	-.063	-.090	-.303
Family Income	.842	.255	-.743	.531	-.169
Age (< 55 = 1)	-.208	.226	-.073	.129	-.165
Sex (Male = 1)	-.009	.198	-.088	-.247*	.141
Education (> HS = 1)	-.015	.080	-.128	-.069	.317*
Housewife (= 1)	-.110	.380***	.058	-.029	-.120
Health Status	.478	.632*	-.059	.645	.724
Health Concerns	.104	-.294	-.552**	-.025	-.141
Income Adequacy	.618	.273	.036	.351	.338
Private Doctor (= 1)	.257	.094	-.118	.322*	-.210
Yrs. with Doctor (< 2 yrs = 1)	.517**	-.026	-.317	.051	.434**
No. Sites Used	.260	-.233	-.030	-.605	-.670
No. of Visits	-1.26 *	-.933*	-1.178*	-.298	-.804
Amb. Care Costs	-.256	-.984***	.159	-.028	.263
Own Car (= 1)	-.195	.128	-.077	-.061	-.207
Travel Time	.152	2.92 ***	.194	.827*	-.040
Time to Appt.	.012	-.094	.865***	.310	-.441
Wait Time in Office	.794**	-.111	1.556***	.578*	.370
F Statistic	.972	5.283***	3.224***	1.624*	1.113
R ²	.096	.367	.261	.151	.109

* Significant at 10%
** Significant at 5%
*** Significant at 1%

all less than 0.4, and only four of 306 independent variable pairs correlated above 0.3.

Results of the five regressions are summarized in Table 6. Independent variables significant at 10 per cent or better in the availability equation suggest that a longer relationship with the physician and more visits in the past 6 months imply greater satisfaction, while longer waiting times in the physician's office decrease satisfaction with availability. As expected, patients with longer travel times are less satisfied with accessibility. In fact, the beta for travel time is three times greater than the next largest variable coefficient. Housewives are less satisfied with accessibility, as are persons with poorer perceived health status. A greater number of ambulatory visits is positively associated with

accessibility satisfaction, as is higher ambulatory care expenditures!

Satisfaction with accommodation is lower for persons having to wait longer for an appointment and having to wait longer in the physician's office. The beta coefficient for "wait time in the office" is substantially greater than that of any other variable in the equation. Patients evidencing greater health concern and those with fewer ambulatory visits in the previous 6-month period also tend to be less satisfied with accommodation.

While variables related to financial cost of care (income adequacy and ambulatory care expenditures) are not significant in the affordability equation, those associated with opportunity cost—travel time and waiting time in the office—are significant and have signs in the expected direction.

In addition to persons with greater opportunity costs, females and patients having a private physician tend to be less satisfied with affordability.

Not surprisingly, a longer relationship with the physician suggests greater satisfaction with acceptability of the provider. Also, persons with less education tend to be more satisfied with this dimension of access.

A separate set of regression analyses was run using the same independent variables as above, but using the summated satisfaction ratings as dependent variables. § R^2 's and sets of significant independent variables were generally consistent with those shown in Table 6, except for the analysis of satisfaction with affordability. With the summated measure of affordability, R^2 was 0.23 instead of 0.15 for the affordability factor regression. Beta coefficients significant at 10 per cent or better in the summated rating equation showed patients with lower perceived income adequacy and higher opportunity costs (those with greater travel times and waiting times and with no private automobile) to be less satisfied with affordability. Also shown to be less satisfied were blacks, persons with lower perceived health status and those having a private physician.

Discussion

The regression results presented above are generally consistent with expectations. For example, travel time is a strong predictor of satisfaction with accessibility; time to get an appointment is predictive of satisfaction with accommodation; and a longer relationship with the physician implies greater satisfaction with availability and acceptability. Having to wait longer in the physician's office negatively influences satisfaction with availability and accommodation, while travel time and waiting

time in the physician's office, together representing opportunity cost of a visit, were shown to influence satisfaction with affordability. As expected, a greater number of visits is associated positively with satisfaction with availability, accessibility and accommodation. And it appears reasonable that educational level would have a stronger influence on satisfaction with acceptability than on other dimensions of access, since education is presumed to influence the values against which "acceptability" is measured.

While Hulka et al.^{26,27} found persons having a private physician to be more satisfied with cost/convenience, the results in Table 6 suggest that this group is less satisfied with the affordability dimension of access. The differing results may be due to different populations studied or to the different nature of the dimensions measured, since cost/convenience encompasses accommodation and accessibility as well as affordability. It may be hypothesized that patients having a private physician resent high fees that are perceived as contribution to the physician's high individual income, while patients using clinics and other less personal sources of care do not make this direct association.

The regression results also indicate that housewives are less satisfied with accessibility than are respondents in other occupational groups. Residences of most persons in the study population are in the suburbs of Rochester, while places of employment and most physicians' offices are in the downtown area. The results suggest that nonemployed females perceive the time or distance to reach care differently than do others in the study population, perhaps because their usual "market basket of travel distances" is less than that of employed persons.

Persons with high health concerns, those who think about their health more than most other people, are shown to be less satisfied than other respondents with the accommodation dimension of access. Ac-

§ Log transforms of the summated scale values were used in these regressions to compensate for the extreme non-normality of the dependent variables.

commodation relates to the “customer service” aspects of access—getting appointments, waiting in the office, telephone consultations—and persons with high health concerns are likely to be more sensitive than others to these factors.

A few of the relationships observed are difficult to explain. Why, for example, does perceived health status influence accessibility satisfaction more than satisfaction with other dimensions of access? Similarly, why do ambulatory care expenditures influence not affordability but accessibility?

In spite of these problems, and in spite of the low R^2 values of two of the equations (availability and acceptability), the regression results must be considered supportive of the construct validity of the proposed access dimensions. The purpose of the regression analysis was to determine if variables found to relate to the different dimensions of access are reasonable in terms of definitions of those dimensions; and results do appear consistent with expectations. Together, results of the factor analysis and regression analysis provide evidence that for the population studied, patients can and do distinguish among availability, accessibility, accommodation, affordability, and acceptability; and that the factor scores and summated ratings do in fact measure aspects of the phenomena with which they are identified.

Summary

The concept of access is central to much of health policy and is referred to extensively in studies of health services utilization and satisfaction. Nevertheless, the concept has been ambiguous and has been used in various ways by researchers and policymakers alike.

It is proposed that access is a measure of the “fit” between characteristics of providers and health services and characteristics and expectations of clients, and that this concept includes five reasonably distinct dimensions: availability, accessibility, ac-

commodation, affordability and acceptability. It was observed that existence of such dimensions is compatible with findings of researchers investigating service utilization and those investigating patient satisfaction toward health care providers and services.

Using data from a survey conducted in Rochester, New York, the same dimensions as those proposed above emerged when a factor analysis was performed on responses to questions dealing with patient satisfaction. Regression analyses, each using one of the five factors as the dependent variable, showed that the factors are generally valid measures of the concepts they are hypothesized to represent. Thus, results of the data analysis provide support to the existence and validity of the access dimensions proposed.

Because few (16) attitudinal questions were used in the factor analysis, our measures may not represent reliable scales for assessing all concepts embodied in each of the dimensions of access. Instruments used in future research should include a larger number of positively and negatively worded questions concerning attitudes toward the five dimensions of access. For example, in addition to the availability questions listed in Figure 1, other questions might assess attitudes concerning degree of difficulty in locating a source of care and in being seen by a provider when care is needed, and necessity for using alternative sources when the patient's usual provider is unavailable. It was noted that variations in access are presumed to influence not only patient satisfaction, but service utilization and provider practice patterns as well. These outcomes are interrelated; system characteristics that affect patient satisfaction negatively may also reduce utilization, either directly or through the mechanism of satisfaction. Low availability of providers may result in demands on the practicing physicians that cannot be met, and this may influence practice patterns of these physicians. Further investi-

gation of the nature of these relationships is needed to determine if the five dimensions of access affect only satisfaction or whether they have independent and measurable effects on consumer behavior and on provider practice patterns that influence utilization. These are the foci of future studies.

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